

Standard form: $a + bi$

$a + bi$

U3-D1 Homework – Imaginary Numbers

Name: Key

Simplify completely. Answers may contain i , but not i^2 .

1. $\sqrt{-64}$
 $= \sqrt{-1 \cdot 64}$
 $= \boxed{8i}$

2. $\sqrt{-175}$
 $= \sqrt{-1 \cdot 25 \cdot 7}$
 $= \boxed{5i\sqrt{7}}$

3. $12\sqrt{-21}$
 $12 \cdot \sqrt{-1 \cdot 21}$
 $= \boxed{12i\sqrt{21}}$

4. $-4\sqrt{-20}$
 $-4 \cdot \sqrt{-1 \cdot 4 \cdot 5}$
 $= -4 \cdot 2 \cdot i\sqrt{5}$
 $= \boxed{-8i\sqrt{5}}$

5. $-7i \cdot 6i$
 $= -7 \cdot 6 \cdot i \cdot i$
 $= -42i^2$
 $= -42(-1) = \boxed{42}$

6. $8i(2 + 5i)$
 $= 16i + 40i^2$
 $= 16i + 40(-1)$
 $= \boxed{-40 + 16i}$

7. $(5 + 4i)(3 - 2i)$ ← FOIL
 $15 - 10i + 12i - 8i^2$
 $= 15 + 2i - 8(-1)$
 $= 15 + 2i + 8$
 $= \boxed{23 + 2i}$

8. $(2 - 9i)(2 + 9i)$ ← FOIL
 $4 + 18i - 18i - 81i^2$
 $4 - 81(-1)$
 $4 + 81$
 $= \boxed{85}$

9. $7i(3 + 5i) + 2(9i - 1)$
 distribute
 $21i + 35i^2 + 18i - 2$
 $21i + 35(-1) + 18i - 2$
 $21i - 35 + 18i - 2$
 $= \boxed{-37 + 39i}$

10. $-i(12 + 3i) + (6 + i)(6 - i)$
 distribute FOIL
 $-12i - 3i^2 + 36 + 6i - 6i - i^2$
 $-12i + 3 + 36 - (-1)$
 $-12i + 39 + 1$
 $= \boxed{40 - 12i}$

Solve.

11. $\frac{3x^2}{3} = \frac{-27}{3}$
 $\sqrt{x^2} = \sqrt{-9}$
 $\boxed{x = 3i}$

12. $x^2 + 40 = 0$
 $-40 \quad -40$
 $\sqrt{x^2} = \sqrt{-40}$
 $x = \sqrt{-1 \cdot 4 \cdot 10}$
 $\boxed{x = 2i\sqrt{10}}$

i^2
 $= (\sqrt{-1})^2$
 $= -1$

Simplify completely.

$$13. \quad \frac{40+6\sqrt{3}}{105} \quad \text{GCF}$$

$$= \frac{20+3\sqrt{3}}{5}$$

$$14. \quad \frac{28-21i}{142} \quad \text{GCF}$$

$$\frac{4-3i}{2}$$

$$15. \quad \frac{-24-\sqrt{60}}{8}$$

$$\frac{-24-\sqrt{15 \cdot 4}}{8}$$

$$\frac{-24-2\sqrt{15}}{84} \quad \leftarrow \text{GCF}$$

$$\boxed{\frac{-12-\sqrt{15}}{4}}$$

$$16. \quad \frac{-30+\sqrt{-25}}{5}$$

$$\frac{-30+5i}{51} \quad \leftarrow \text{GCF}$$

$$\boxed{-6+i}$$

17. Solve using the quadratic formula. Simplify completely.

$$x^2 - 10x + 22 = 0 \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-10) \pm \sqrt{(-10)^2 + 4(1)(22)}}{2(1)}$$

$$x = \frac{10 \pm \sqrt{100 + 88}}{2}$$

$$x = \frac{10 \pm \sqrt{188}}{2}$$

$$x = \frac{10 \pm \sqrt{4 \cdot 47}}{2}$$

$$x = \frac{10 \pm 2\sqrt{47}}{21} \quad \leftarrow \text{GCF} = 5 \pm \sqrt{47}$$

Factor completely.

$$18. \quad 16x^{13} - 54x$$

$$2x(8x^{12} - 27) \quad \leftarrow a^3 - b^3$$

$$\boxed{2x(2x^4 - 3)(4x^8 + 6x^4 + 9)}$$

$$19. \quad 4x^2 - 5x - 21 \quad \frac{-84}{-12 \mid +7}$$

$$4x^2 - 12x + 7x - 21$$

$$4x(x-3) + 7(x-3)$$

$$\boxed{(4x+7)(x-3)}$$